



5.8 Further integration: substitution and by parts (HL)

Calculus · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Integration by substitution reverses the chain rule: for $\int f(g(x))g'(x)dx$, let $u=g(x)$, $du=g'(x)dx$, transforming the integral into $\int f(u)du$. Integration by parts reverses the product rule: $\int u dv = uv - \int v du$, useful for integrands that are a product of functions (e.g. $x \cdot e^x$, $x \cdot \ln x$), typically choosing u to be the factor that simplifies upon differentiation.

Worked example: follow the reasoning

Find $\int 2x(x^2+1)^4 dx$ using substitution.

Let $u=x^2+1$, $du=2x dx$.

$$\int 2x(x^2+1)^4 dx = \int u^4 du = \frac{u^5}{5} + C = \frac{(x^2+1)^5}{5} + C.$$

Quick check · multiple choice

Using integration by parts on $\int x \cdot e^x dx$, a suitable choice is:

A $u=e^x$, $dv=x dx$

B $u=x$, $dv=e^x dx$

C $u=xe^x$, $dv=dx$

D Substitution is required instead

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 (a) Use substitution to find $\int x\sqrt{x^2+9} dx$.
(b) Use integration by parts to find $\int x \cdot \ln(x) dx$.

[8]

Final answer: _____



Answers and explanation

Question 1 [8]

(a) Let $u=x^2+9$, $du=2x \, dx$, so $x \, dx = du/2$.

$$\int x\sqrt{x^2+9}dx = \int \sqrt{u} \cdot (du/2) = (1/2)\int u^{1/2}du = (1/2)(2/3)u^{3/2}+C = (1/3)(x^2+9)^{3/2}+C.$$

(b) Let $u=\ln(x)$, $dv=x \, dx$, so $du=(1/x)dx$, $v=x^2/2$.

$$\int x \ln(x)dx = (x^2/2)\ln(x) - \int (x^2/2)(1/x)dx = (x^2/2)\ln(x) - \int (x/2)dx = (x^2/2)\ln(x) - x^2/4 + C.$$

Quick check answer

B. Choosing $u=x$ (which simplifies to 1 upon differentiation) and $dv=e^x \, dx$ (easily integrated) makes integration by parts straightforward.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics: AA subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.